

Cell culture dishes are often coated in cationic polymers, which have been shown to help tissues with negatively charged membranes and membrane proteins adhere to the dish. Which of the following explains why poly-D-lysine appears to be more effective at keeping cells attached than poly-L-lysine?

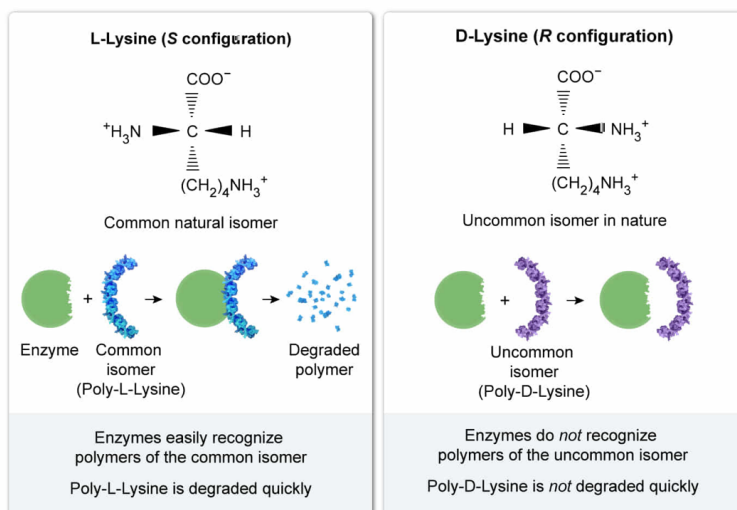
- ✔ ☒ A. Poly-D-lysine is less likely to be broken down by most proteases than poly-L-lysine. [54%]
- ☐ B. Poly-D-lysine can be broken down into a more efficient fuel source than poly-L-lysine, leading to healthier cultured cells. [3%]
- ☐ C. The average charge of poly-D-lysine is more positive than poly-L-lysine. [14%]
- ☐ D. The structure of poly-D-lysine more closely mimics natural cellular ligands than poly-L-lysine. [27%]

Correct

53% answered correctly

Explanation:

D-Amino acids are uncommon in nature, and are less likely to be recognized by proteases



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Stereoisomers are molecules that have the same composition and connectedness of atoms but with **different spatial orientations** of

atoms around at least one **stereocenter**. A **chiral center** is a stereocenter that has a **nonsuperimposable mirror image** and is commonly formed when an sp^3 -hybridized carbon is attached to four different substituents, as is the case at the α -carbon of most **amino acids**. Molecules that differ from each other at all chiral centers are known as **enantiomers**. Enantiomers **share the same physical and chemical properties** (with the exception of spatial orientation), including melting point, boiling point, and acidity (ie, pK_a value).

However, *biochemical* differences between enantiomers can arise

because the **enzymes that act as biological catalysts are themselves chiral**. Most enzymes that act on peptides, such as proteases, have evolved to **recognize and catalyze reactions with peptides composed of the more common amino acid enantiomers**, which are by convention called the **L-amino acids**.

Therefore, proteases secreted by the cultured cells or by contaminating microbes are more likely to recognize and degrade peptides composed of the common L-amino acids, releasing them from the bottom of the culture dish. In contrast, proteases are **unlikely to degrade peptides composed of D-amino acids**, and therefore these peptides remain bound to the bottom of the dish and allow cells to adhere.

(Choice B) Because D-amino acids are less common than L-amino acids, they are *less* likely than L-amino acids to be used as efficient fuel sources.

(Choice C) L-lysine and D-lysine are enantiomers, as are their polymers (poly-L-lysine and poly-D-lysine). Enantiomers share the same pK_a value and will show the same average charge at the same pH.

(Choice D) The question states that the adhesion of cells is mediated by charge interactions rather than specific binding to

ligands. Because D-amino acids are less common than L-amino acids, it is less likely that proteins will specifically recognize D-amino acids over L-amino acids.

Educational objective:

All amino acids (except glycine) are chiral and exist most commonly in nature as L-enantiomers. Peptides composed of D-enantiomers are less common in nature and less likely to be recognized by enzymes.

Time spent:0 sec

Subject:
Biochemistry

QID:403545

System:
Amino Acids and Proteins

Acid-sensing ion channels (ASICs) are ligand-gated ion channels found in the central and peripheral nervous systems. Each ASIC subunit contains a pair of glutamate residues positioned close to each other within the ligand-binding pocket. At a pH of 7.4, the ionized glutamate residues repel each other, but under acidic conditions the protonated glutamate residues form stable hydrogen bonds. This bonding initiates a series of conformational changes resulting in the opening of the channel and the passage of sodium ions. As such, ASICs enhance the signaling response from neurotransmitters released by acidic synaptic vesicles, but they can also be overactivated during inflammation or ischemia, leading to neuronal damage.

In an effort to minimize neuronal damage, scientists worked to discover antibody-based therapeutics to neutralize ASIC activity during inflammatory events like strokes. Human ASIC proteins were recombinantly expressed, solubilized in detergent, purified, and then inserted into nanodiscs. Nanodiscs are composed of small lipid-bilayer discs that can hold a single copy of an integral membrane protein of interest.

The phospholipid tails at the edges of the disc are stabilized by an amphipathic α -helical protein known as membrane scaffold protein (MSP). Amphipathic α -helices consist of both hydrophobic and hydrophilic residues. Each hydrophobic residue is separated from other hydrophobic residues by two or three hydrophilic residues. The use of nanodisc-encapsulated proteins ensured that the purified ASICs maintained native structure and function, which was confirmed by structural studies.

A phage display library consisting of 100 billion members—each encoding and expressing a different antibody—was then used to discover high-affinity antibodies against ASIC. Phages that bound purified ASIC were collected and used to infect *Escherichia coli*, where the phage genome encoding the antibody was amplified and subsequently isolated. The selected antibodies were expressed and biochemically tested for binding and unbinding rate constants against the nanodisc-encapsulated ASIC receptors at a pH of 7.4 (Table 1).

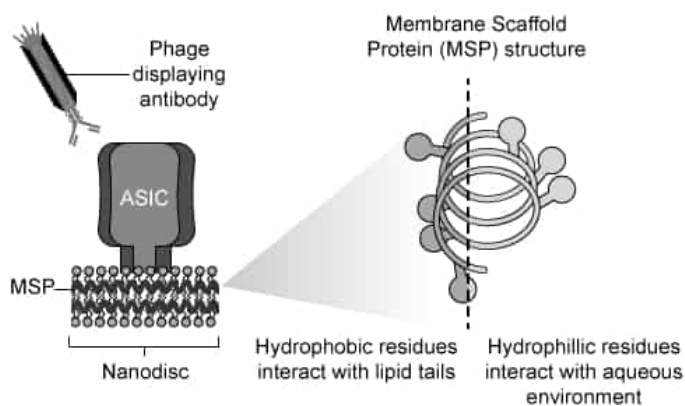


Figure 1 Overview of the phage display system used to discover high-affinity antibodies.

Table 1 Binding and Unbinding Rate Constants for Four Antibody Candidates

Antibody	$k_{\text{binding}} \text{ (M}^{-1}\text{s}^{-1}\text{)}$	$k_{\text{unbinding}} \text{ (s}^{-1}\text{)}$
Ab1-IgG	2.1×10^5	8.4×10^{-5}
Ab3-IgG	1.1×10^7	8.9×10^{-1}
Ab4-IgG	5.3×10^6	4.8×10^{-4}
Ab6-IgG	3.0×10^5	2.4×10^{-3}

Based on the positions of the glutamate residues in the ASIC ligand-binding pocket, how do the pK_a values of these residues most likely compare to the pK_a of free, monomeric glutamate side chains in solution?

- ☐ A. Ligand-binding pocket glutamate residues have a lower pK_a than free glutamate. [30%]
- ✔ ☒ B. Ligand-binding pocket glutamate residues have a higher pK_a

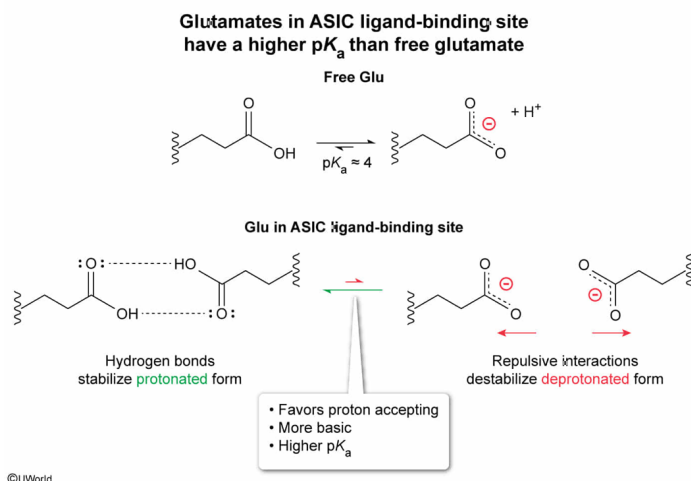
than free glutamate. [39%]

- ☐ C. Ligand-binding pocket glutamate residues have the same pK_a as free glutamate. [23%]
- ☐ D. This cannot be determined without further information. [6%]

Correct

40% answered correctly

Explanation:



The **acid dissociation equilibrium constant** K_a is a measure of the acidity of an ionizable molecule or **functional group**. It can be determined by calculating the **reaction quotient** with **equilibrium concentrations** for the acid dissociation reaction:



$$K_{eq, \text{ acid dissociation}} = K_a = \frac{[H^+]_{eq} [A^-]_{eq}}{[HA]_{eq}}$$

Taking the negative base-10 logarithm of the K_a value transforms it into the pK_a value. Similar to pH, a *lower* pK_a value indicates a *more acidic*

functional group, and a **higher pK_a** indicates a **less acidic** functional group. Based on the [Henderson-Hasselbalch equation](#), when $pH = pK_a$, the concentrations of A^- and HA are equal.

The [p \$K_a\$ of free glutamate](#) is approximately 4. However, the values of K_a and pK_a can change if **local effects** alter the **stabilities** of the products and reactants, such as through [inductive effects](#) or through interactions with nearby polar or charged species. The passage states that the nearby ionized glutamate residues (ie, A^-) **repel** each other in the ASIC ligand-binding pocket; conversely, the protonated form of the residues (ie, HA) participate in **stabilizing hydrogen bonds**. Therefore, equilibrium shifts **away from the deprotonated form** (products) and **toward the protonated form** (reactants). This shift corresponds to an **increased pK_a** (ie, a **less** acidic molecule that is **more likely** to be **protonated**).

(Choice A) A lower pK_a would only occur if the ionized form were *stabilized* or the *protonated* form destabilized.

(Choice C) The destabilization of the deprotonated form of glutamate, and the stabilization of its protonated form, cause changes in the pK_a values of those residues.

(Choice D) The effect on equilibrium constant values can be estimated based on knowledge of the stabilities of the products and reactants of the reaction, which are given in the passage.

Educational objective:

The pK_a value is a measure of the acidity of a functional group.

Destabilization of the deprotonated form or stabilization of the protonated form leads to a decrease in acidity and an increase in pK_a .

Time spent: 0 sec

Subject:
Biochemistry

QID: 403546

System:
Amino Acids and Proteins

Acid-sensing ion channels (ASICs) are ligand-gated ion channels found in the central and peripheral nervous systems. Each ASIC subunit contains a pair of glutamate residues positioned close to each other within the ligand-binding pocket. At a pH of 7.4, the ionized glutamate residues repel each other, but under acidic conditions the protonated glutamate residues form stable hydrogen bonds. This bonding initiates a series of conformational changes resulting in the opening of the channel and the passage of sodium ions. As such, ASICs enhance the signaling response from neurotransmitters released by acidic synaptic vesicles, but they can also be overactivated during inflammation or ischemia, leading to neuronal damage.

In an effort to minimize neuronal damage, scientists worked to discover antibody-based therapeutics to neutralize ASIC activity during inflammatory events like strokes. Human ASIC proteins were recombinantly expressed, solubilized in detergent, purified, and then inserted into nanodiscs. Nanodiscs are composed of small lipid-bilayer discs that can hold a single copy of an integral membrane protein of interest.

The phospholipid tails at the edges of the disc are stabilized by an amphipathic alpha-helical protein known as membrane scaffold protein (MSP). Amphipathic α -helices consist of both hydrophobic and hydrophilic residues. Each hydrophobic residue is separated from other hydrophobic residues by two or three hydrophilic residues. The use of nanodisc-encapsulated proteins ensured that the purified ASICs maintained native structure and function, which was confirmed by structural studies.

A phage display library consisting of 100 billion members—each encoding and expressing a different antibody—was then used to discover high-affinity antibodies against ASIC. Phages that bound purified ASIC were collected and used to infect *Escherichia coli*, where the phage genome encoding the antibody was amplified and subsequently isolated. The selected antibodies were expressed and biochemically tested for binding and unbinding rate constants against the nanodisc-encapsulated ASIC receptors at a pH of 7.4 (Table 1).

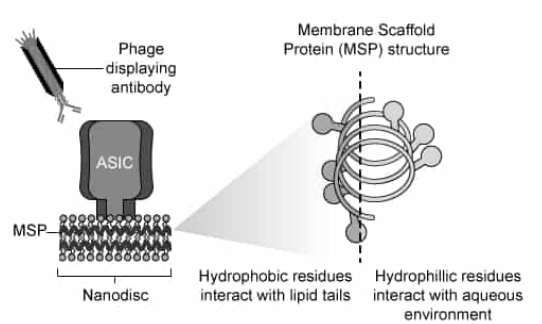


Figure 1 Overview of the phage display system used to discover high-affinity antibodies.

Table 1 Binding and Unbinding Rate Constants for Four Antibody Candidates

Antibody	k_{binding} ($\text{M}^{-1}\text{s}^{-1}$)	$k_{\text{unbinding}}$ (s^{-1})
Ab1-IgG	2.1×10^5	8.4×10^{-5}
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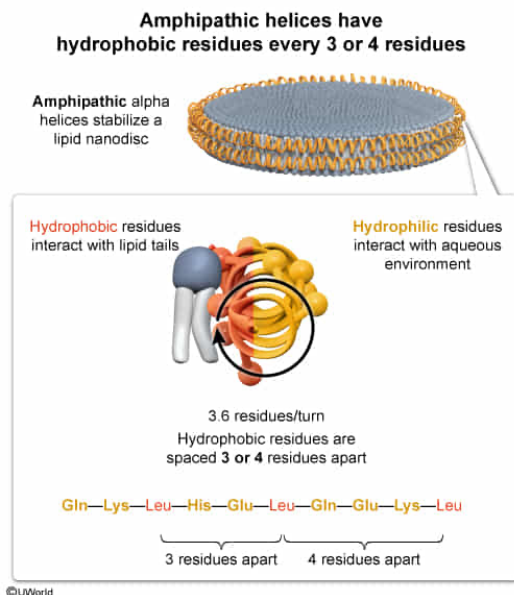
Based on the passage, which of the following peptide sequences is most likely to be found in the alpha-helical region of MSP?

- ☐ A. Ile–Glu–Val–Ser–Leu–Arg–Val–Gln–Phe–Lys [16%]
- ✓ ☒ B. Gln–Lys–Leu–His–Glu–Leu–Gln–Glu–Lys–Leu [58%]
- ☐ C. Arg–Glu–Asp–Ser–Arg–Lys–Asn–Asp–Thr–Asp [14%]
- ☐ D. Leu–Phe–Ala–Val–Ile–Trp–Val–Ile–Leu–Leu [11%]

Correct

58% answered correctly

Explanation:



Alpha helices are important protein **secondary structure** elements that possess the following characteristics:

- Alpha helices are **right-handed**. When looking from one end of the helix, the peptide backbone proceeds in a counterclockwise spiral toward the opposite end.
- Alpha helices have **defined dimensions**. There are **3.6 amino acid residues per 360° turn**, and the distance from one point on the helix to another point 360° away is 5.4 Å (**angstroms**). Therefore, a 1.5 Å rise exists along the axis per residue (5.4 Å / 3.6 residues).
- The backbone carbonyl oxygen of the residue at position i forms a **hydrogen bond** with the amide N–H of residue $i + 4$ to maintain the structure of the helix.
- The **side chains all point outward** from the axis of the helix. The surface properties of the helix are fully determined by the side chains of its residues.

The passage states that MSP stabilizes nanodiscs by forming **amphipathic** alpha helices, meaning that it must have both hydrophilic and hydrophobic side chains in some pattern. More specifically, the passage states that each hydrophobic residue is separated from other hydrophobic residues by two or three hydrophilic residues. This is depicted in Figure 1, which shows that the hydrophobic residues all align on one side of the alpha helix to interact with the

lipid tails whereas the other side is exposed to the aqueous buffer. Of the given options, only the sequence of Choice B shows both hydrophobic and hydrophilic residues with the appropriate spacing between hydrophobic residues.

(Choice A) This option displays *alternating* hydrophobic and hydrophilic residues. This pattern would be expected if MSP were an **amphipathic beta-pleated sheet**, not an alpha helix.

(Choice C) This option has all hydrophilic residues. It is not amphipathic.

(Choice D) This option has all hydrophobic residues. It is not amphipathic.

Educational objective:

Alpha helices are protein secondary structure elements that have 3.6 residues per turn. To maintain a hydrophobic face of an amphipathic alpha helix, hydrophobic residues must be spaced 3 or 4 residues apart.

Time spent:0 sec

Subject:
Biochemistry

QID:403547

System:
Amino Acids and Proteins

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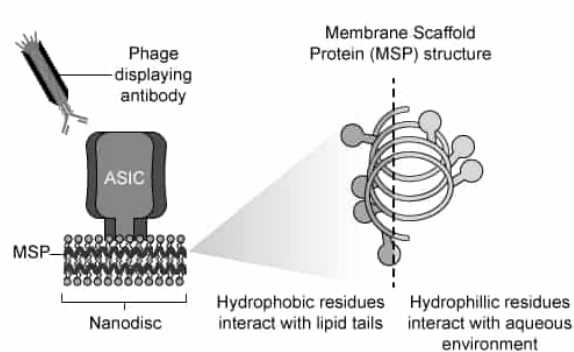


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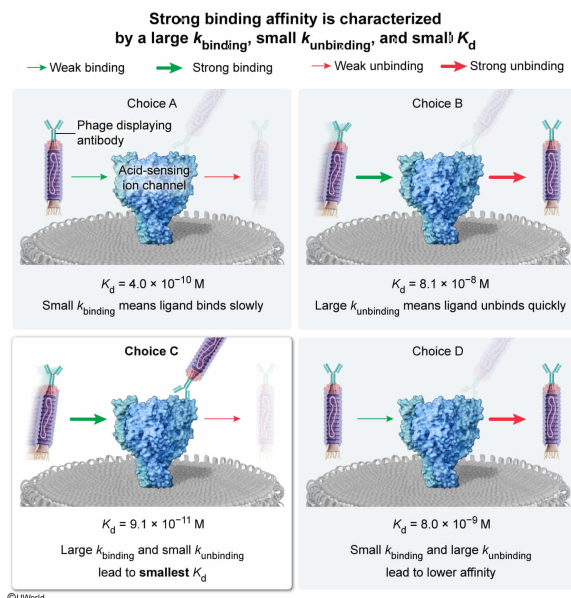
Based on both the k_{binding} and $k_{\text{unbinding}}$ values given in Table 1, which of the antibodies displays the strongest overall binding affinity (K_d) toward ASIC?

- ☐ A. Ab1-IgG [24%]
☐ B. Ab3-IgG [36%]
☒ C. Ab4-IgG [32%]
☐ D. Ab6-IgG [6%]

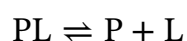
Correct

32% answered correctly

Explanation:



The **ligand dissociation equilibrium constant** K_d is a measure of the affinity of a ligand L for its receptor protein P to form a PL complex (eg, an antibody toward its antigen forming an antibody-antigen complex). K_d can be determined by calculating the **reaction quotient** with **equilibrium concentrations** for the ligand dissociation reaction:



$$K_{\text{eq, ligand dissociation}} = K_d = \frac{[P]_{\text{eq}}[L]_{\text{eq}}}{[PL]_{\text{eq}}}$$

Equilibrium constants like K_d can also be **derived** from the **rate constants** of the forward reaction (ie, $k_{\text{unbinding}}$) and the reverse reaction (ie, k_{binding}). Because the forward and reverse rates are **equal at equilibrium**, the K_d expression is the ratio of $k_{\text{unbinding}}$ to k_{binding} , or:

$$K_d = \frac{k_{\text{unbinding}}}{k_{\text{binding}}}$$

This relationship is also sometimes described as:

$$K_d = \frac{k_{\text{off}}}{k_{\text{on}}}$$

From this relationship, the individual K_d values can be calculated and compared. To facilitate calculations without a calculator, these values can also be **approximated to one significant figure**—values that differ by an order of magnitude or greater, or by greater than two units in the least significant place value, can be compared without more precise calculations.

From these data, **Ab4-IgG** shows the **smallest K_d** ($4.8 \times 10^{-4} / 5.3 \times 10^6 = 9.1 \times 10^{-11} \text{ M}$) and therefore displays the strongest overall binding affinity and the least net tendency to dissociate.

(Choice A) Although Ab1-IgG has the smallest unbinding rate constant, it also has a small binding rate constant, indicating that binding is slow, unlikely, and of lower affinity than Ab4-IgG.

(Choice B) Although Ab3-IgG has the largest binding rate constant, it also has a large unbinding rate constant, indicating that bound antibodies will rapidly disassociate, and therefore are of low affinity.

(Choice D) Ab6-IgG has both a smaller binding constant and a larger unbinding constant than Ab4-IgG, indicating that it is of lower affinity than Ab4-IgG.

Educational objective:

The dissociation constant K_d can be calculated either with equilibrium concentrations

$\left(\frac{[P]_{\text{eq}}[L]_{\text{eq}}}{[PL]_{\text{eq}}} \right)$ or with rate constants $\left(\frac{k_{\text{unbinding}}}{k_{\text{binding}}} \right)$. Smaller K_d values indicate stronger overall binding affinity.

Time spent: 0 sec
Subject:
Biochemistry

QID: 403548
System:
Amino Acids and Proteins

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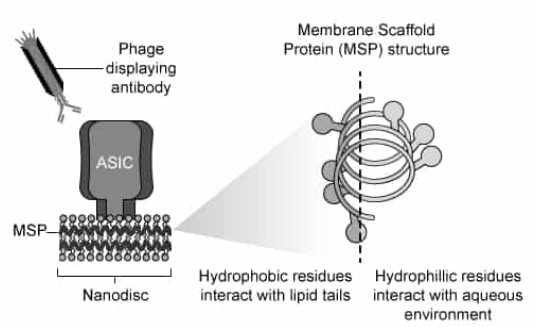


Figure 1 Overview of the phage display system used to discover high-affinity antibodies.

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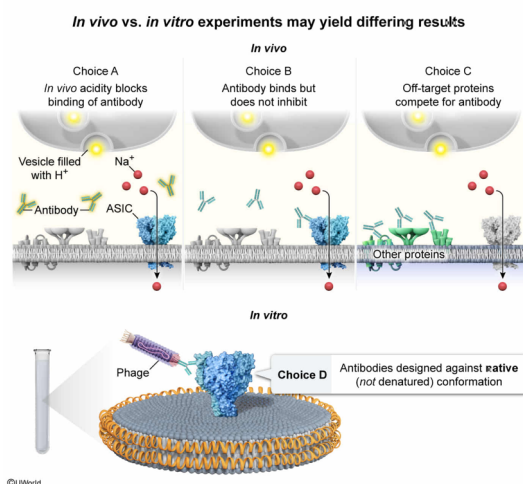
In a follow-up study, the antibody candidates were tested *in vivo*. The researchers found that the most protective antibody *in vivo* was not the antibody with the strongest affinity for ASIC *in vitro*. All the following could explain this observation EXCEPT:

- ☐ A. the antibodies may act differently at the *in vivo* pH than at the *in vitro* pH. [19%]
- ☐ B. the antibody with the strongest binding affinity *in vitro* may bind, but not inhibit, ASIC *in vivo*. [20%]
- ☐ C. the antibodies may bind off-target ligands *in vivo* that were not present *in vitro*. [24%]
- ✔ ☒ D. the antibodies with the strongest binding affinity *in vitro* may preferentially recognize denatured proteins. [35%]

Correct

34% answered correctly

Explanation:



Physiological, *in vivo* verification of biochemical experiments is an important complement to *in vitro* experiments such as binding assays. The complexities of living organisms can often cause unexpected variables to arise such as unexpected ligand interactions, compensatory mechanisms, and altered drug half-lives. Off-target effects from nonspecific binding (eg, binding to proteins other than ASIC) can also affect the efficacy of a therapeutics *in vivo* versus *in vitro* (**Choice C**). For this reason, antibodies that are tested only for binding in one *in vitro* assay might yield different results than expected when tested *in vivo* for effects on functionality.

The passage states that acid-sensing ion channels respond to **acidic** conditions. Pathologic acidity, which can be caused by ischemia due to the buildup of **lactic acid**, can **overactivate** ASICs, leading to neurotoxicity and cell death. The tested antibodies are intended to **neutralize** ASICs under these conditions, preventing their overactivation to **minimize damage**. However,

because the passage states that the antibodies were tested at pH 7.4 only, their *in vitro* binding efficacy may not be valid at the acidic pH present in pathological states (**Choice A**). Furthermore, binding ability as measured through K_d does not give any information about inhibitory ability. If the antibodies bind ASIC **but do not inhibit** its activity, they are not expected to be protective (**Choice B**).

The question asks for the reasoning that does *not* explain why an antibody with weaker binding gives better protection. The ability of the antibodies to recognize native-conformation ASIC is

unlikely to be an issue **for this particular** experiment. Although an important consideration in other assays (eg, choosing an antibody for a western blot following SDS-PAGE versus live-cell immunofluorescence), the passage indicates that the antibody development process and the binding assay were both performed on **native-conformation proteins**. Functional proteins *in vivo* are also expected to be in their native conformation; therefore, the antibodies in this experiment are expected to recognize their target either *in vitro* or *in vivo*, and preferentially binding to denatured proteins would **not** explain the observed difference.

Educational objective:

Physiological, *in vivo* verification of biochemical assays complements *in vitro* studies. Effective neutralizing antibodies intended to inhibit their target should show decreased activity of that target in an *in vivo* experiment.

Time spent:0 sec

Subject:
Biochemistry

QID:403549

System:
Amino Acids and Proteins